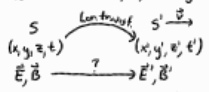


- E/M P in Lorentzova transformacija

imamo 2 sistemu S, S', S' se giblje s hitrostjo blizu c v x-smerni



- Lorentzova transf. koordinat

$$\begin{matrix} x' = \gamma(x - \beta ct) \\ y' = y \\ z' = z \\ ct' = \gamma(ct - \beta x) \end{matrix} \quad \begin{matrix} \beta = \frac{v}{c} \\ \gamma = \frac{1}{\sqrt{1 - \beta^2}} \end{matrix}$$

- Lorentzova transformacija $\vec{E}$ in $\vec{B}$

skrajšava se oblika Maxwellovih enačb:

$$\begin{matrix} \nabla \cdot \vec{E}' = \rho \\ \nabla \times \vec{E}' = -\frac{\partial \vec{B}'}{\partial t'} \end{matrix} \quad \begin{matrix} \nabla \cdot \vec{B}' = 0 \\ \nabla \times \vec{B}' = \frac{1}{c} \frac{\partial \vec{E}'}{\partial t'} \end{matrix}$$

$$\begin{aligned} \rightarrow \frac{\partial E'_x}{\partial x'} + \frac{\partial E'_y}{\partial y'} + \frac{\partial E'_z}{\partial z'} &= 0 \rightarrow \text{operabilimo} \quad \frac{\partial}{\partial x'} = \gamma \left(\frac{\partial}{\partial x} + \beta \frac{\partial}{\partial ct} \right) \rightarrow \begin{matrix} E'_x = E_x \\ E'_y = \gamma(E_y + v B_z) \\ E'_z = \gamma(E_z - v B_y) \\ B'_x = B_x \\ B'_y = \gamma(B_y - \frac{v}{c^2} E_z) \\ B'_z = \gamma(B_z + \frac{v}{c^2} E_y) \end{matrix} \\ \frac{\partial E'_y}{\partial y'} - \frac{\partial E'_z}{\partial z'} &= -\frac{\partial B'_x}{\partial t'} \quad \frac{\partial}{\partial y'} = \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z'} = \frac{\partial}{\partial z} \\ \frac{\partial E'_z}{\partial z'} - \frac{\partial E'_x}{\partial x'} &= -\frac{\partial B'_y}{\partial t'} \quad \frac{\partial}{\partial ct'} = \gamma \left(\frac{\partial}{\partial ct} + \beta \frac{\partial}{\partial x} \right) \\ \frac{\partial E'_x}{\partial x'} - \frac{\partial E'_y}{\partial y'} &= -\frac{\partial B'_z}{\partial t'} \end{aligned}$$

Lorentzova transf. za $\vec{E}$ in $\vec{B}$

- posledice

- $\vec{E} \cdot \vec{B} = E_x B_x + E_y B_y + E_z B_z = \dots = \vec{E}' \cdot \vec{B}'$
- Skalarni produkt je invarianten - kotni med poljem se ohranjajo
- $E^2 - c^2 B^2 = E'^2 - c^2 B'^2$

- prostor Minkovskega

- $x_\mu = (x, y, z, ct)$ - vektor
- invariants spredaj: kovariantni vektor
- $x^\mu = (x, y, z, -ct)$
- invariants zgoraj: kontravariantni vektor
- $x_\mu x^\mu = x^2 + y^2 + z^2 - (ct)^2 = x'_\mu x'^\mu$ invarianten

- štruktura gostote toka

- gostoto nabija in LT
- Naboj mora biti invarianten na LT
- $q = \int \rho dV = \int j^\mu dV^\mu$
- $\int \rho dx dy dz = \int \rho' dx' dy' dz' = \int j'^0 dx' dy' dz' \Rightarrow \boxed{\rho' = \gamma \rho}$
- $\rightarrow \frac{\rho}{\gamma}$ je invarianten na LT
- $j_\mu = \frac{q}{V} u_\mu = \frac{q}{V} (\gamma \vec{v}, \gamma c) = \left(\frac{j}{\gamma}, \rho c \right)$
- $\leftarrow$ kotni hitrosti

$$j^\mu = \begin{pmatrix} j \\ \rho c \end{pmatrix} \quad j_\mu j^\mu = j^2 - \rho^2 c^2 \quad \frac{\partial}{\partial x_\mu} \rightarrow \boxed{\nabla^2 - \frac{\partial^2}{\partial t^2}} \text{ kont. en.}$$

- LT gostote toka

$$\begin{aligned} j'_0 &= \gamma(j_0 - \beta j_1) \\ j'_1 &= j_1 \\ j'_2 &= j_2 \\ j'_3 &= j_3 \\ \rho' c &= \gamma(\rho c - \beta j_x) \end{aligned}$$

- vektor EM potenciala

- Riemann-Lorentzova en.
- $\square^2 \Psi = \square^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \Psi}{\partial t^2} = -\frac{\rho}{\epsilon_0}$
- $\square^2 \vec{A} = \square^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{j}$
- $\square^2 = \square_\mu \square^\mu$ D'Alembertov operator
- $(\partial_\mu, \frac{\partial}{\partial ct})$
- $A_\mu = (\vec{A}, \Psi/c)$ - vektor EM potenciala

$$\begin{aligned} \square A_\mu &= -\mu_0 j_\mu \\ \square A^\mu &= -\mu_0 j^\mu \end{aligned}$$

- LT EM potenciala

$$\begin{aligned} A'_0 &= \gamma(A_0 - \beta A_1) \\ A'_1 &= A_1 \\ A'_2 &= A_2 \\ A'_3 &= A_3 \\ \Psi'/c &= \gamma(\Psi/c - \beta A_x) \end{aligned}$$

- Schwarzschildova invarianta v rel. obl.

$$\int (-\rho \Psi + \vec{j} \cdot \vec{A}) dV dt = \int A_\mu j^\mu d^4x_\mu = \int A^\mu_{,\nu} j_\mu d^4x_\mu$$

- Kovariantni tenzor EMP

$$\begin{aligned} E_x &= -\frac{\partial \Psi}{\partial x} = -c \frac{\partial A_0}{\partial x^1} + c \frac{\partial A_1}{\partial x^0} \\ B_x &= \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} = \frac{\partial A_3}{\partial x^2} - \frac{\partial A_2}{\partial x^3} \end{aligned} \quad \left. \vphantom{\begin{aligned} E_x \\ B_x \end{aligned}} \right\} \text{ izmenja se kot 4D tenzor}$$

- vedemo tenzor EMP:

$$\begin{aligned} F_{\mu\nu} &= \frac{\partial A_\nu}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^\nu} \text{ Antisimetričen} \\ F_{\mu\nu} &= \begin{bmatrix} 0 & B_3 & -B_2 & -E_1/c \\ -B_3 & 0 & B_1 & -E_2/c \\ B_2 & -B_1 & 0 & -E_3/c \\ E_1/c & E_2/c & E_3/c & 0 \end{bmatrix} \quad \begin{matrix} \text{v 4D EMP postane} \\ \text{eno polje} \end{matrix} \end{aligned}$$

- kontravariantni tenzor EMP

$$\begin{aligned} F^{\mu\nu} &= \frac{\partial A^\nu}{\partial x^\mu} - \frac{\partial A^\mu}{\partial x^\nu} \\ F^{\mu\nu} &= \begin{bmatrix} 0 & B_3 & -B_2 & E_1/c \\ -B_3 & 0 & B_1 & E_2/c \\ B_2 & -B_1 & 0 & E_3/c \\ -E_1/c & -E_2/c & -E_3/c & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} F_{\mu\nu} F^{\mu\nu} &= 2(B^2 - \frac{E^2}{c^2}) \quad (\text{Energijna polja}) \\ \det F_{\mu\nu} &= \frac{1}{c^4} (\vec{B} \cdot \vec{E})^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} F_{\mu\nu} F^{\mu\nu} \\ \det F_{\mu\nu} \end{aligned}} \right\} \text{ invarianti}$$

- Kovariantna akcija EMP

$$S = \int \left(\frac{1}{2} \epsilon_{\mu\nu} E^\mu E^\nu - \frac{1}{2} B^2 - \rho\phi + \vec{j} \cdot \vec{A} \right) d^3x dt =$$

$$\frac{ie}{c} \int \left[\frac{1}{2} F_{\mu\nu} F^{\mu\nu} + j_\mu A^\mu \right] d^4x$$